```python
import tensorflow as tf
from tensorflow.keras import layers
import numpy as np
import matplotlib.pyplot as plt
```

```
2026-03-18 12:25:29.797968: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to
register cuFFT factory: Attempting to register factory for plugin
cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are
written to STDERR
E0000 00:00:1773836729.956651      55 cuda_dnn.cc:8579] Unable to
register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
E0000 00:00:1773836730.003777      55 cuda_blas.cc:1407] Unable to
register cuBLAS factory: Attempting to register factory for plugin
cuBLAS when one has already been registered
W0000 00:00:1773836730.375521      55 computation_placer.cc:177]
computation placer already registered. Please check linkage and avoid
linking the same target more than once.
W0000 00:00:1773836730.375559      55 computation_placer.cc:177]
computation placer already registered. Please check linkage and avoid
linking the same target more than once.
W0000 00:00:1773836730.375562      55 computation_placer.cc:177]
computation placer already registered. Please check linkage and avoid
linking the same target more than once.
W0000 00:00:1773836730.375564      55 computation_placer.cc:177]
computation placer already registered. Please check linkage and avoid
linking the same target more than once.
```

```python
(x_train, _), (_, _) = tf.keras.datasets.cifar10.load_data()

x_train = x_train.astype('float32')
x_train = (x_train - 127.5) / 127.5  # Normalize to [-1, 1]

BUFFER_SIZE = 60000
BATCH_SIZE = 128

dataset =
tf.data.Dataset.from_tensor_slices(x_train).shuffle(BUFFER_SIZE).batch
(BATCH_SIZE)
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-
python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 2s 0us/step

I0000 00:00:1773836758.055219      55 gpu_device.cc:2019] Created
device /job:localhost/replica:0/task:0/device:GPU:0 with 13757 MB
memory:  -> device: 0, name: Tesla T4, pci bus id: 0000:00:04.0,
compute capability: 7.5
```

```
I0000 00:00:1773836758.061112      55 gpu_device.cc:2019] Created
device /job:localhost/replica:0/task:0/device:GPU:1 with 13757 MB
memory:  -> device: 1, name: Tesla T4, pci bus id: 0000:00:05.0,
compute capability: 7.5
```

```
latent_dim = 100

def build_generator():
    model = tf.keras.Sequential()

    model.add(layers.Dense(8*8*256, use_bias=False,
input_shape=(latent_dim,)))
    model.add(layers.BatchNormalization())
    model.add(layers.LeakyReLU())

    model.add(layers.Reshape((8, 8, 256)))

    model.add(layers.Conv2DTranspose(128, (5,5), strides=(1,1),
padding='same', use_bias=False))
    model.add(layers.BatchNormalization())
    model.add(layers.LeakyReLU())

    model.add(layers.Conv2DTranspose(64, (5,5), strides=(2,2),
padding='same', use_bias=False))
    model.add(layers.BatchNormalization())
    model.add(layers.LeakyReLU())

    model.add(layers.Conv2DTranspose(3, (5,5), strides=(2,2),
padding='same', use_bias=False, activation='tanh'))

    return model

generator = build_generator()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
dense.py:93: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an
`Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)
```

```
def build_discriminator():
    model = tf.keras.Sequential()

    model.add(layers.Conv2D(64, (5,5), strides=(2,2), padding='same',
input_shape=[32,32,3]))
    model.add(layers.LeakyReLU())
    model.add(layers.Dropout(0.3))

    model.add(layers.Conv2D(128, (5,5), strides=(2,2),
padding='same'))
```

```
    model.add(layers.LeakyReLU())
    model.add(layers.Dropout(0.3))

    model.add(layers.Flatten())
    model.add(layers.Dense(1))

    return model

discriminator = build_discriminator()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/
convolutional/base_conv.py:113: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)
```

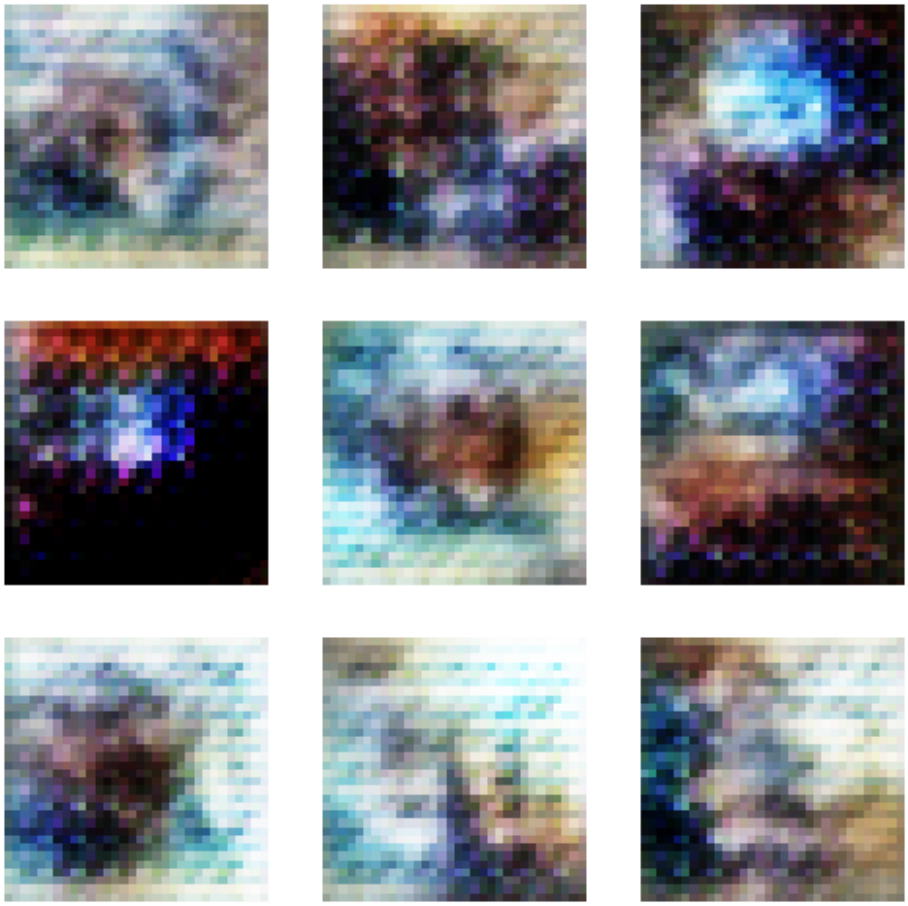
```
cross_entropy = tf.keras.losses.BinaryCrossentropy(from_logits=True)

def generator_loss(fake_output):
    return cross_entropy(tf.ones_like(fake_output), fake_output)

def discriminator_loss(real_output, fake_output):
    real_loss = cross_entropy(tf.ones_like(real_output), real_output)
    fake_loss = cross_entropy(tf.zeros_like(fake_output), fake_output)
    return real_loss + fake_loss

generator_optimizer = tf.keras.optimizers.Adam(1e-4)
discriminator_optimizer = tf.keras.optimizers.Adam(1e-4)

@tf.function
def train_step(images):
    noise = tf.random.normal([BATCH_SIZE, latent_dim])

    with tf.GradientTape() as gen_tape, tf.GradientTape() as
disc_tape:
        generated_images = generator(noise, training=True)

        real_output = discriminator(images, training=True)
        fake_output = discriminator(generated_images, training=True)

        gen_loss = generator_loss(fake_output)
        disc_loss = discriminator_loss(real_output, fake_output)

    gradients_gen = gen_tape.gradient(gen_loss,
generator.trainable_variables)
    gradients_disc = disc_tape.gradient(disc_loss,
discriminator.trainable_variables)

    generator_optimizer.apply_gradients(zip(gradients_gen,
```

```
generator.trainable_variables))
    discriminator_optimizer.apply_gradients(zip(gradients_disc,
discriminator.trainable_variables))

EPOCHS = 20

for epoch in range(EPOCHS):
    for image_batch in dataset:
        train_step(image_batch)
    print(f"Epoch {epoch+1} completed")
```

```
E0000 00:00:1773836790.543893      55 meta_optimizer.cc:967] layout
failed: INVALID_ARGUMENT: Size of values 0 does not match size of
permutation 4 @ fanin shape
insequential_1_2/dropout_1/stateless_dropout/SelectV2-2-
TransposeNHWCToNCHW-LayoutOptimizer
I0000 00:00:1773836790.774674     130 cuda_dnn.cc:529] Loaded cuDNN
version 91002

Epoch 1 completed
Epoch 2 completed
Epoch 3 completed
Epoch 4 completed
Epoch 5 completed
Epoch 6 completed
Epoch 7 completed
Epoch 8 completed
Epoch 9 completed
Epoch 10 completed
Epoch 11 completed
Epoch 12 completed
Epoch 13 completed
Epoch 14 completed
Epoch 15 completed
Epoch 16 completed
Epoch 17 completed
Epoch 18 completed
Epoch 19 completed
Epoch 20 completed
```

```
def generate_images(num_images=9):
    noise = tf.random.normal([num_images, latent_dim])
    generated_images = generator(noise, training=False)

    generated_images = (generated_images + 1) / 2

    plt.figure(figsize=(6,6))
    for i in range(num_images):
        plt.subplot(3,3,i+1)
        plt.imshow(generated_images[i])
```

```
        plt.axis('off')
    plt.show()

generate_images()
```



```
def interpolate(z1, z2, steps=10):
    z = [(1 - t) * z1 + t * z2 for t in np.linspace(0, 1, steps)]
    z = tf.stack(z)

    images = generator(z, training=False)
    images = (images + 1) / 2

    plt.figure(figsize=(10,2))
    for i in range(steps):
        plt.subplot(1, steps, i+1)
        plt.imshow(images[i])
        plt.axis('off')
    plt.show()

z1 = tf.random.normal([latent_dim])
z2 = tf.random.normal([latent_dim])
```

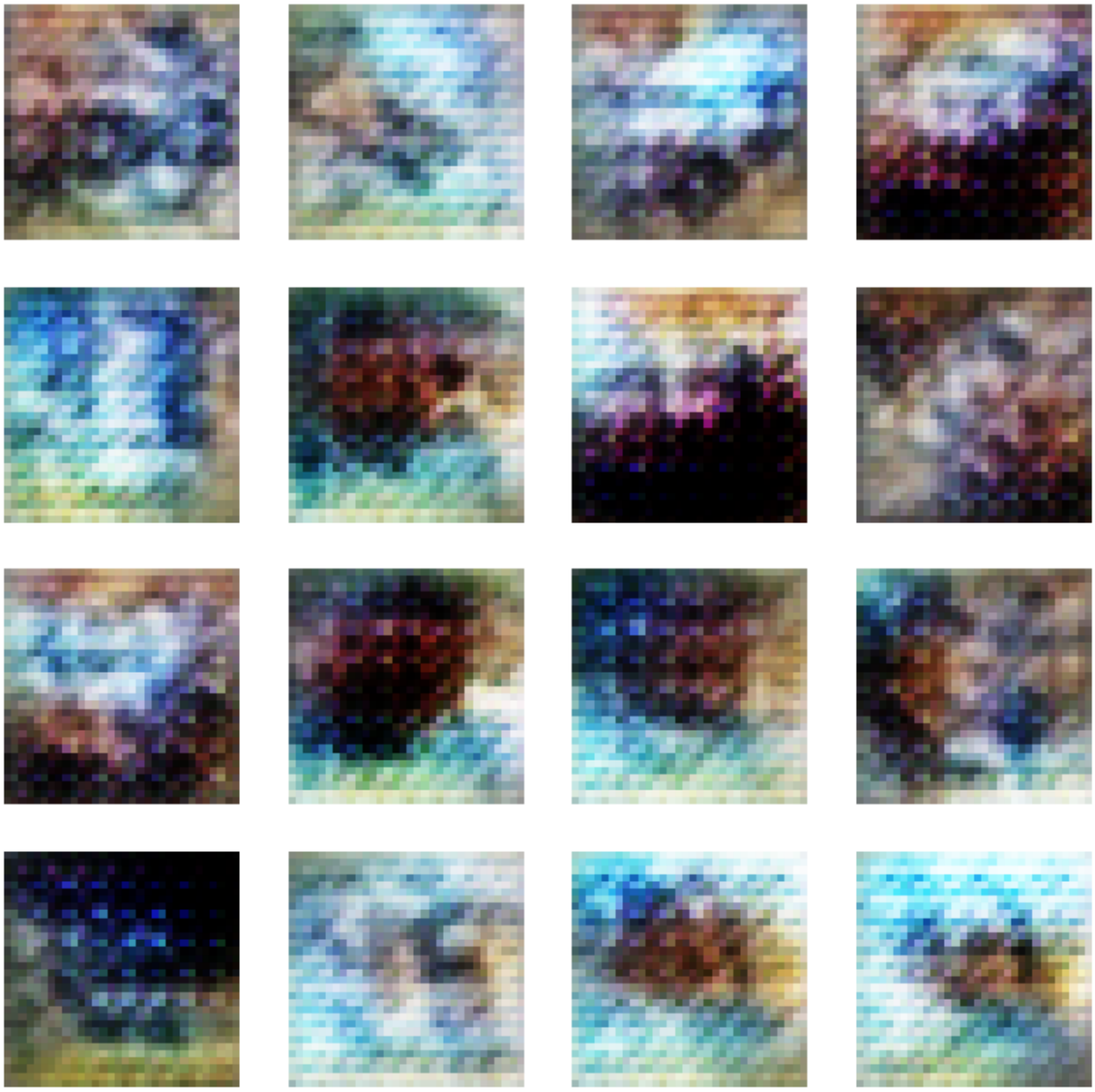
```
interpolate(z1, z2)
```

```
generator.save("dcgan_generator.h5")
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`.
```

```
def visualize_latent_diversity():
    noise = tf.random.normal([16, latent_dim])
    images = generator(noise, training=False)
    images = (images + 1) / 2

    plt.figure(figsize=(8,8))
    for i in range(16):
        plt.subplot(4,4,i+1)
        plt.imshow(images[i])
        plt.axis('off')
    plt.suptitle("Different Latent Vectors → Different Outputs",
fontsize=14)
    plt.show()

visualize_latent_diversity()
```

Different Latent Vectors → Different Outputs



```
def smooth_interpolation():
    z1 = tf.random.normal([latent_dim])
    z2 = tf.random.normal([latent_dim])

    steps = 12
    interpolated = [(1-t)*z1 + t*z2 for t in np.linspace(0,1,steps)]
    interpolated = tf.stack(interpolated)
```

```
    images = generator(interpolated, training=False)
    images = (images + 1) / 2

    plt.figure(figsize=(12,2))
    for i in range(steps):
        plt.subplot(1, steps, i+1)
        plt.imshow(images[i])
        plt.axis('off')
    plt.suptitle("Latent Space Interpolation", fontsize=14)
    plt.show()

smooth_interpolation()
```

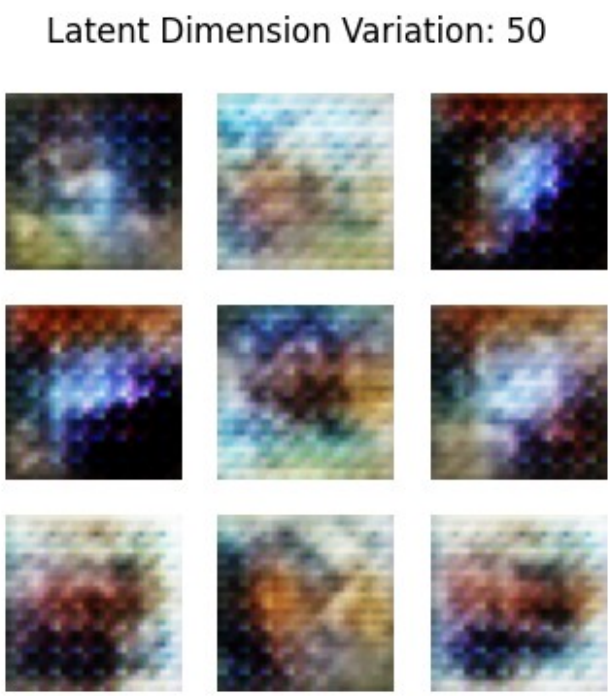
Latent Space Interpolation

```
fixed_noise = tf.random.normal([1, latent_dim])

def test_stability():
    plt.figure(figsize=(6,2))
    for i in range(5):
        img = generator(fixed_noise, training=False)
        img = (img + 1) / 2
        plt.subplot(1,5,i+1)
        plt.imshow(img[0])
        plt.axis('off')
    plt.suptitle("Same Latent Vector → Same Output", fontsize=14)
    plt.show()

test_stability()
```

Same Latent Vector → Same Output

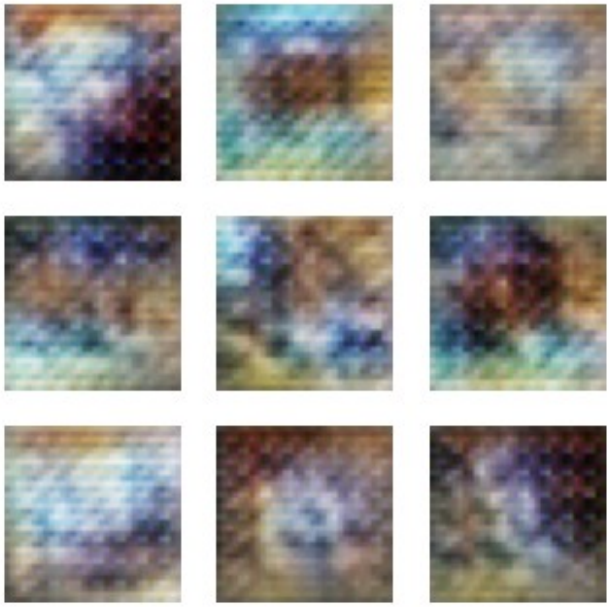
```
latent_dims = [50, 100, 200]

for dim in latent_dims:
    noise = tf.random.normal([9, dim])
```

```
    # resize noise if needed (simple trick)
    noise = tf.image.resize(tf.reshape(noise, (9, dim, 1, 1)),
(latent_dim,1))
    noise = tf.reshape(noise, (9, latent_dim))

    images = generator(noise, training=False)
    images = (images + 1) / 2

    plt.figure(figsize=(4,4))
    for i in range(9):
        plt.subplot(3,3,i+1)
        plt.imshow(images[i])
        plt.axis('off')
    plt.suptitle(f"Latent Dimension Variation: {dim}", fontsize=12)
    plt.show()
```



Latent Dimension Variation: 100

Latent Dimension Variation: 200



```
num_classes = 10

def build_cgan_generator():
    noise = layers.Input(shape=(latent_dim,))
    label = layers.Input(shape=(1,), dtype='int32')
```

```
    label_embedding = layers.Embedding(num_classes, latent_dim)(label)
    label_embedding = layers.Flatten()(label_embedding)

    model_input = layers.multiply([noise, label_embedding])

    x = layers.Dense(8*8*256)(model_input)
    x = layers.Reshape((8,8,256))(x)
    x = layers.Conv2DTranspose(128, 4, strides=2, padding='same')(x)
    x = layers.ReLU()(x)
    x = layers.Conv2DTranspose(3, 4, strides=2, padding='same',
activation='tanh')(x)

    return tf.keras.Model([noise, label], x)

cgan_generator = build_cgan_generator()

def generate_diverse_images():
    noise = tf.random.normal([10, latent_dim])
    images = generator(noise, training=False)
    images = (images + 1) / 2

    plt.figure(figsize=(10,2))
    for i in range(10):
        plt.subplot(1,10,i+1)
        plt.imshow(images[i])
        plt.axis('off')
    plt.suptitle("DCGAN: Diverse Artistic Outputs", fontsize=14)
    plt.show()

generate_diverse_images()
```

DCGAN: Diverse Artistic Outputs